MINGWEI XU

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ABOUT ME

My research interests in **LLM post-training**, **reinforcement learning**, **multimodal reasoning** and **agentic AI**. My work spans model alignment and reasoning, multimodal test-time improvement and **AI for scientific discovery**. I aim to build reliable and adaptive AI systems that can reason over complex evidence.

EDUCATION EXPERIENCE

University of Washington 2025.09 – Present
M.S. in Electrical and Computer Engineering Seattle, WA, USA
• **GPA:** 3.8/4.0
• **Selected Coursework:** Algorithms and Data Structures, Machine Learning, AI for Robot Control

Zhejiang University of Technology 2021.09 – 2025.06
B.S. in Automation Hangzhou, Zhejiang, China
• **GPA:** 3.6/4.0; **Rank:** 3/118
• **Honors:** Outstanding Graduate; Zhejiang Provincial Government Scholarship (3×); First-Class Outstanding Student Scholarship (2×); Innovation and Entrepreneurship Scholarship (2×)

INTERNSHIP EXPERIENCE

Zhejiang Lab 2025.07–2025.09, 2026.07–2026.09
Research Intern, LLM Post-Training and Scientific Discovery
• **LLM Post-Training:** Investigated post-training and alignment strategies for **Qwen-14B** and **32B** on an **8×A100 GPU** cluster; conducted reinforcement learning experiments (**PPO**, **GRPO**) with **LoRA** across model scales; designed **benchmark evaluations**, **ablation studies** and pipelines to analyze training and prompt.
• **GeoGPT & Scientific Discovery:** Conducted **pretraining** and **instruction tuning** of GeoGPT for geoscience reasoning; curated and optimized domain-specific training datasets for post-training.
• Developed a **scientific Agent** for automated hypothesis generation and research in protein–disease interactions, integrating **RAG**, evidence retrieval and **PPI networks** for biomarker discovery, validation and interpretation.

Hangzhou Closeli Technology Co., Ltd. 2024.05–2024.07
Automation Software Engineer, Embodied Robot Control
• Developed embedded control software for a multi-joint robotic arm on **STM32**, implementing real-time sensor communication (**UART or SPI or I2C**), trajectory planning and **closed-loop PID** control modules.
• Designed a sensor-feedback pipeline integrating **force or torque** and **proximity signals** for adaptive motion execution, enabling real-time grasping force adjustment and obstacle avoidance.
• Collaborated on testing, debugging latency and **control loop stability** across multi-sensor configurations.

RESEARCH EXPERIENCE

Positive-Only Policy Optimization for LLM Reasoning (POPO) [Github] 2025.11–2026.09
• Existing **RLVR** relies on negative updates that destabilize training under sparse binary rewards. We proposed **POPO**, a positive-only framework eliminating negative updates while maintaining stable on-policy optimization.
• Designed a **self-distillation** architecture combining bounded importance weighting, EMA anchoring, representation-space alignment and entropy regularization, preventing reward hacking and collapse without negative rollouts.
• Evaluated POPO on **Qwen2.5-Math-1.5B** and **7B**, **DeepSeek-R1-Distill-1.5B** and **7B**, Llama-3.1-8B and DeepSeek-Math-7B, achieving **36.67% Pass@8** on AIME 2025 with Qwen2.5-Math-7B, vs. **30.00%** with GRPO.
• Paper accepted to the **ICONIP 2026** and **ICML Workshop DEMO 2026**.

AV-Phys Bench: Physics in Joint Audio-Video Generation [Website] 2026.03–2026.09
• No existing **benchmark** evaluates whether **joint audio-video models** respect **physical commonsense**. We contributed to designing a **physics-grounded** evaluation framework addressing this gap.
• Developed the **benchmark** with **321 prompts** across Steady State, Event Transition and Environment Transition, with **Anti-Physics** stress tests and fine-grained rubrics for visual, audio and cross-modal consistency.
• Built **AV-Phys Agent**, a **ReAct-style MLLM** evaluator integrating deterministic acoustic tools to automate rubric-based assessment, achieving rankings **aligned with human evaluation** across seven generation models.
• Submitted to the **NeurIPS 2026** and **rebuttal scores: 5/5/4**.

Knowledge-Informed Disease Protein Biomarker Discovery 2026.07–2026.09

• Biomarker selection optimizes prediction accuracy, ignoring biological relevance. We developed a **knowledge-informed framework** that balances predictive power with **protein–protein interaction (PPI)** evidence.
• Proposed a **PPI-Pareto multi-objective genetic algorithm** with **dynamic backward pruning** to optimize AUC, PPI quality, evaluation stability, and feature sparsity under a **predictive non-inferiority constraint**.
• Evaluated in **18 experiments** on Parkinson's (47,572 participants) and T2D (46,472 participants), improving **PPI quality** and **holdout AUC** across in all six comparisons while reducing biomarker panels by **12–17%**.

Databricks Grounded Reasoning Cup (Google DeepMind Sponsored) [Website] 2026.02–2026.06

• Designed a **grounded reasoning Agent** with the **Google Antigravity SDK**, using deterministic two-tier routing to classify query scope, external-data requirements, temporal range and output format, then estimate complexity across lookup, aggregation or CAGR and statistical-analysis tasks to determine the reasoning strategy.
• Built an **grounding reasoning pipeline** that assembles **Gemini-embedding-2** and **FAISS** retrieval summaries, multimodal figure or table evidence, optional web and era-aware context, then selects tier-specific prompts and executes a single Agent with **Python** and **MCP corpus tools** for evidence-grounded multi-step reasoning.
• Represented the **University of Washington** in the Google track, using **Gemini models** across six rounds, competing against MIT, Yale, and Columbia on Databricks OfficeQA grounded reasoning over U.S. Treasury Bulletins (1939–2025); ranked **2nd** among the four Google-sponsored teams, outperforming MIT and Columbia.

FinTech Agentic QA [Github] 2026.01–2026.03

• Developed **single-agent and multi-agent financial QA systems**, with an Orchestrator coordinating Market, Fundamentals and Sentiment agents, integrating financial tools, **SQLite retrieval** and **LLM-as-a-Judge**.
• Single-agent GPT-4o-mini achieved **64.4% vs. 35.6%** on simple questions; multi-agent GPT-4o reached **46.7%** on complex questions and outperformed the single-agent, highlighting collaboration benefits on harder tasks.

SELECTED PUBLICATIONS

• Z. Cui, X. Liu, H. Fang, **M. Xu**, J. Liu, Z. Xu, W. Pian, S. Deng, F. Du, C. Ge, and Y. Tian, "Do Joint Audio-Video Generation Models Understand Physics?" *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. (CCF-A; **Under Review**; Rebuttal Scores: 5/5/4)
• **M. Xu** and H. Fang, "Improve Reasoning Ability by Reinforcing Only from Positive Rollouts," *ICML 2026 Workshop DEMO*, 2026. (**Accepted**)
• **M. Xu** and H. Fang, "Improve Reasoning Ability by Self-Distillation and On-Policy Optimization of Positive Rollouts," *International Conference on Neural Information Processing (ICONIP)*, 2026. (CCF-C; **Accepted**)
• **M. Xu**, H. Deng, H. Xu, and Y. Zhang, "Time Series Forecasting and Correlation Analysis of the New Energy Electric Vehicle Industry Using ARIMA Algorithm and LSTM Neural Networks," *International Conference on Data Analytics, Computing and Artificial Intelligence (ICDACAI)*, 2024. (EI; **Accepted**)
• J. Li, H. Yang, **M. Xu**, Y. Wu, X. Shou, Z. Huang, Y. Hao, et al., "Task-Dependent Cortical Oscillatory Dynamics in Functional Constipation," *Sensors*, 2026. (JCR Q2; IF: 3.5; **Published**)
• B. Tu, Z. Yu, S. Jin, Y. Zhang, **M. Xu**, M. Zhang, Y. Jiang, X. Zhang, J. Weng, and X. Han, "Do Large Language Models Suffer from Cognitive Overload? A Benchmark and Orchestration Framework," *International Conference on Intelligent Computing (ICIC)*, 2026. (CCF-C; **Accepted**)

HONORS AND AWARDS

• **Meritorious Winner**, Mathematical Contest / Interdisciplinary Contest in Modeling (MCM/ICM), 2025.
• **National First Prize & 2nd Place**, National Undergraduate Intelligent Car Competition, 2024.
• **National Third Prize**, China Robot Competition and RoboCup China Open – Underwater Track, 2024.
• **National Third Prize**, National Undergraduate Electronics Design Contest, 2023.
• **National Second Prize**, National Undergraduate Mathematics Competition, 2022.

SPECIALTIES AND SKILLS

• **Programming language:** Python (Pytorch, Tensorflow, Sklearn, NumPy, Scipy), MATLAB, C++, C, Java
• **Tools:** Git, Linux, Altium Designer, FSDP, vLLM, Adobe Illustrator, Latex, Github, Microsoft Office
• **Platforms:** STM32, Raspberry Pi, Infineon TC264, Embedded Systems
• **Language:** Mandarin (native), English (fluent)
• **Reviewer:** Conference on Language Modeling (**COLM 2026**); International Conference on Neural Information Processing (**ICONIP 2026**); International Conference on Artificial Intelligence in Education (**AIED 2026**).
• **Patents:** **M. Xu,**, "Automatic Warehouse Discharging Device for Bottle-Cultivated Edible Fungi," Chinese Utility Model Patent No. ZL 2023 2 3189247.6; **M. Xu,**, "Enoki Mushroom Light-Shield Cover Removal Device," Chinese Utility Model Patent No. ZL 2023 2 3189248.0.